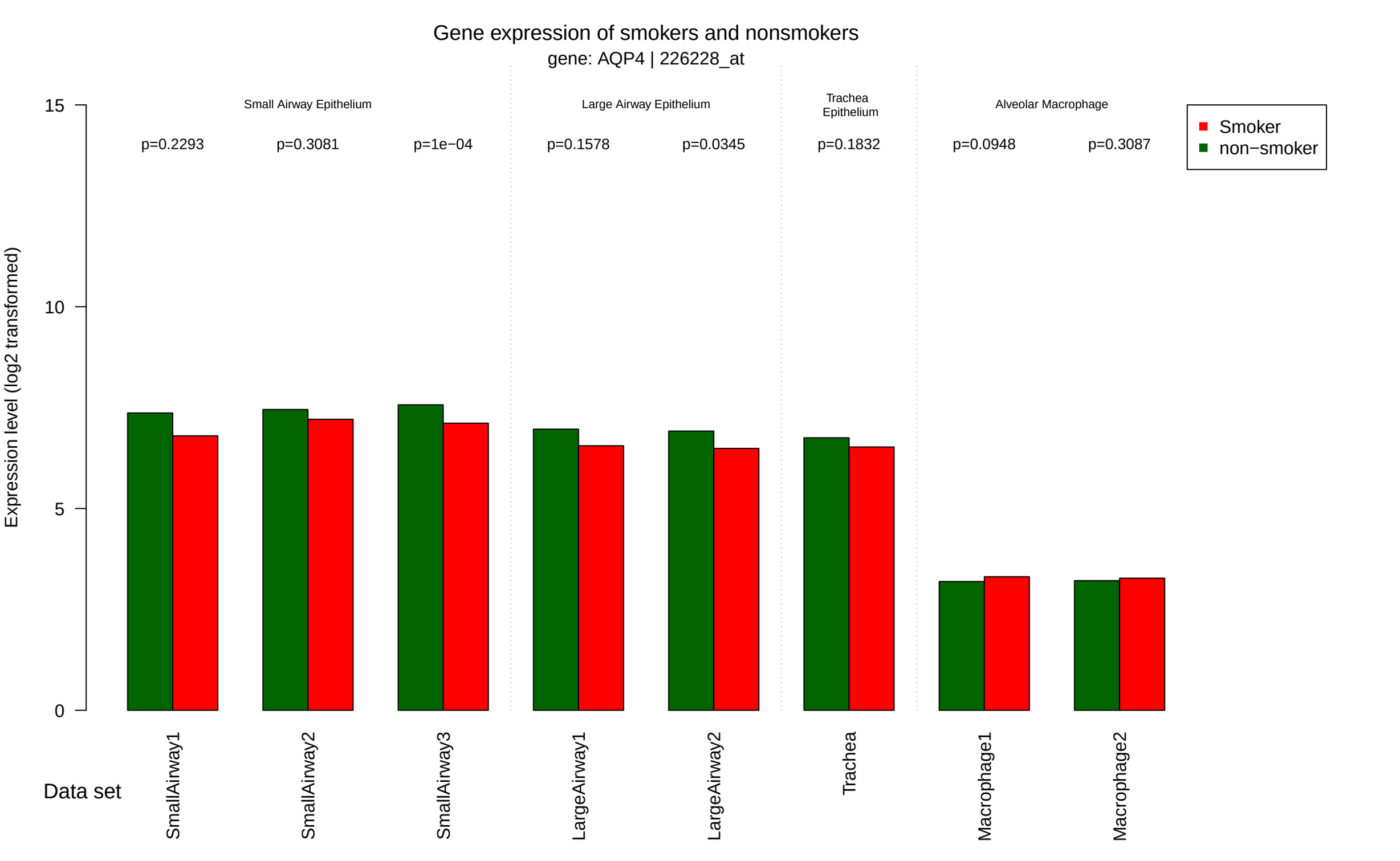
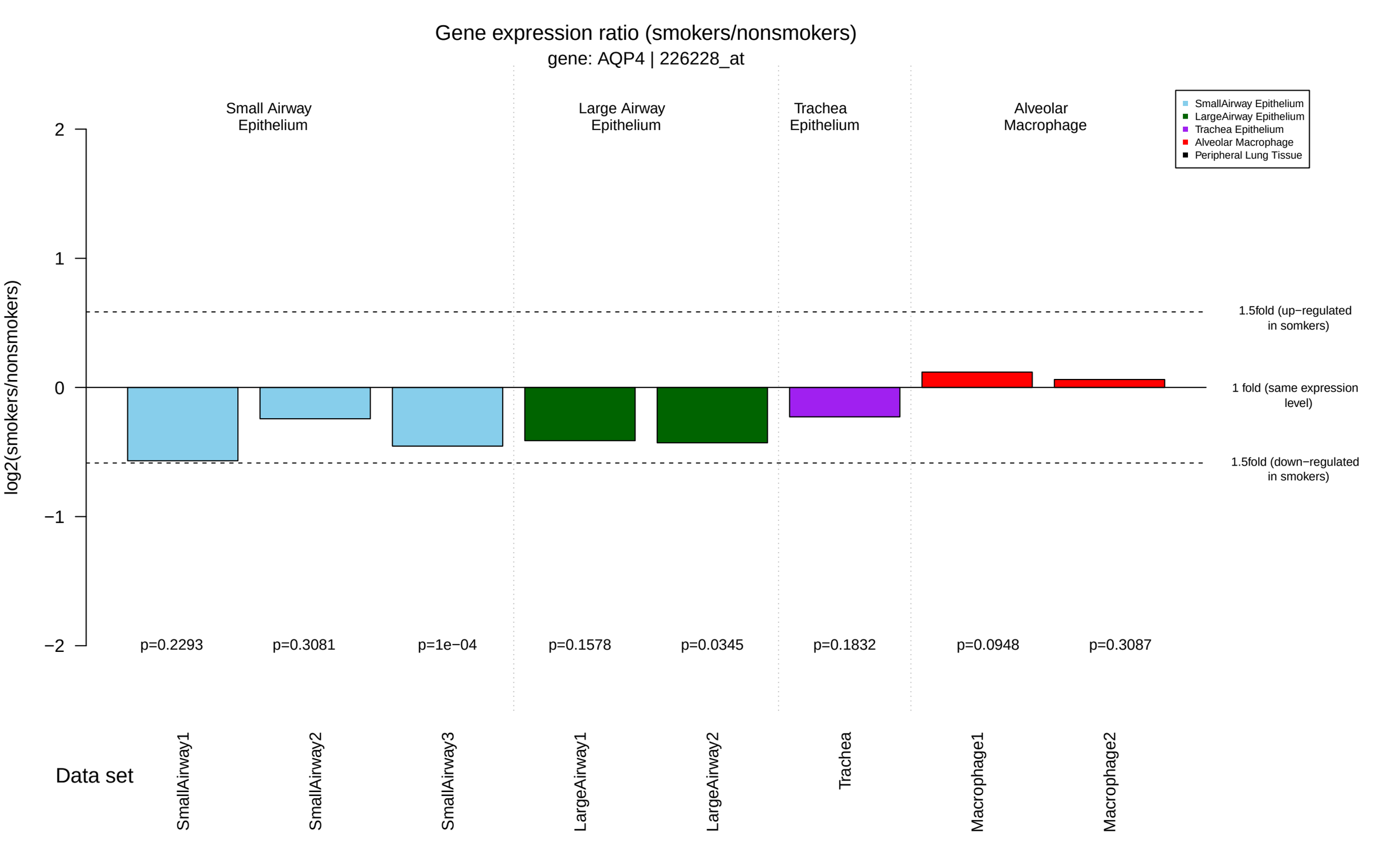
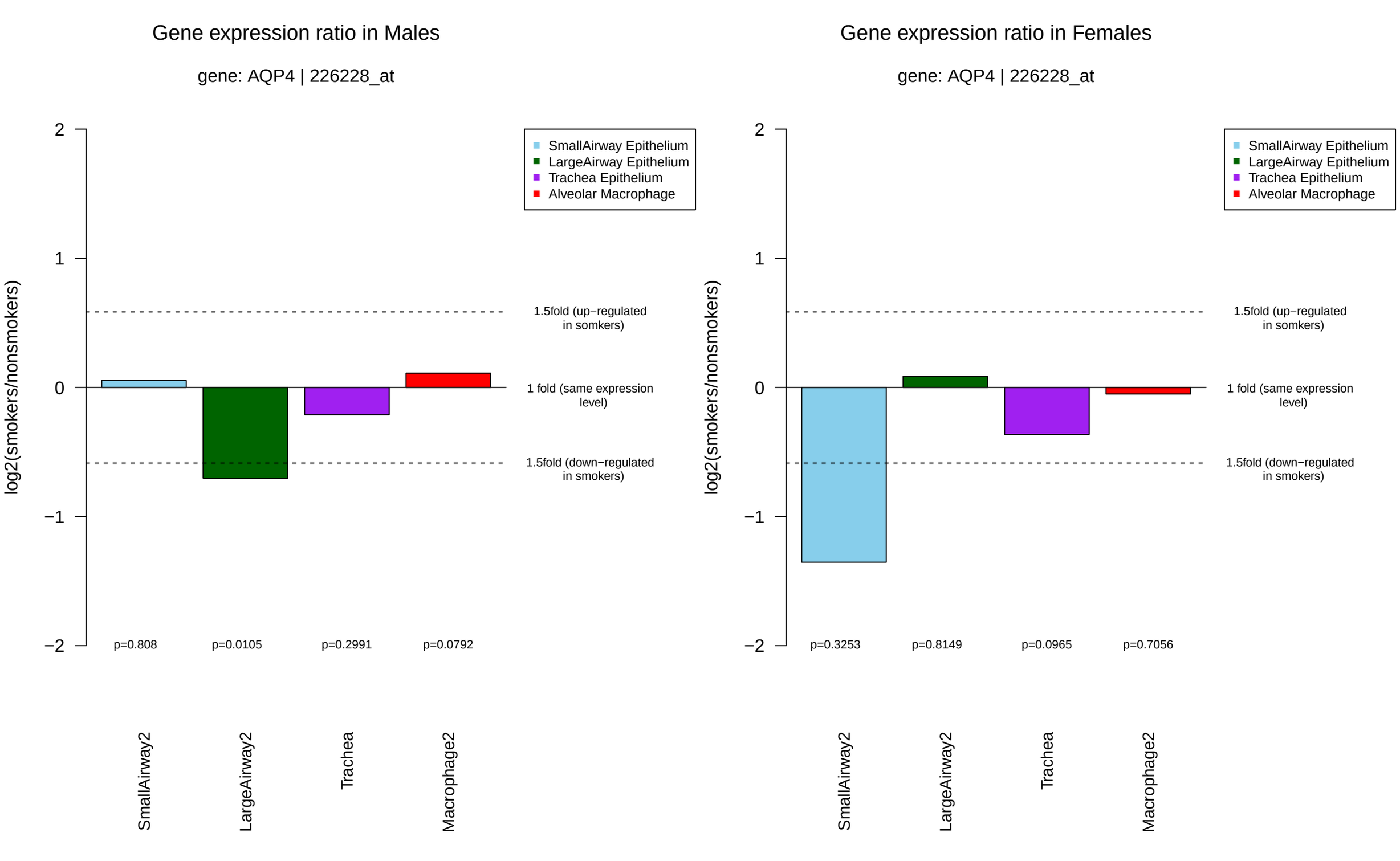
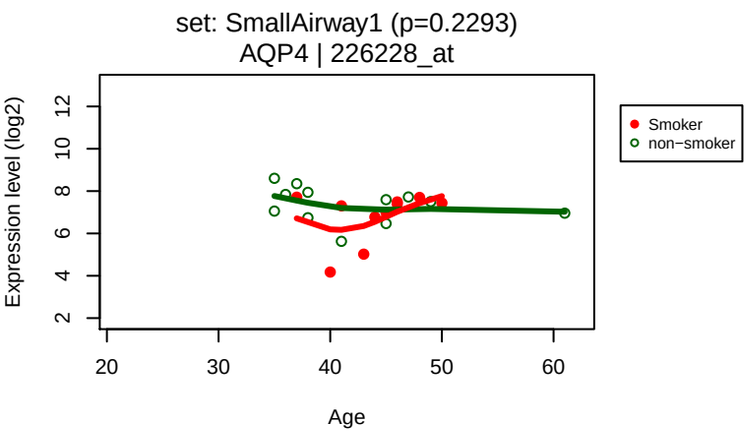
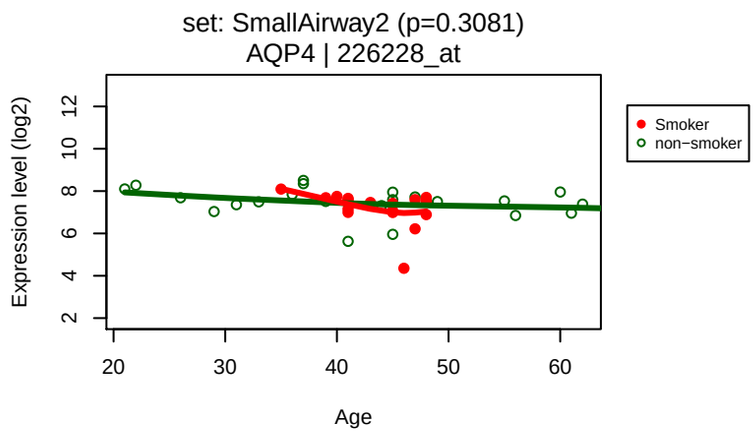
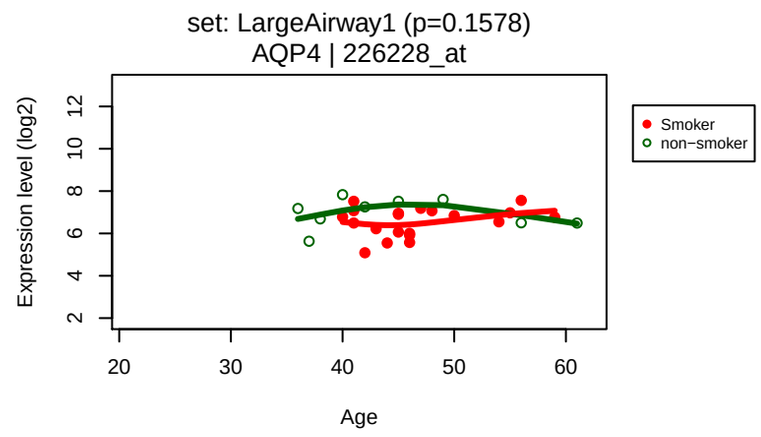

Gene expression of smokers and nonsmokers
gene: AQP4 | 226228_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2293
p=0.3081
p=1e−04
p=0.1578
p=0.0345
p=0.1832
p=0.0948
p=0.3087
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

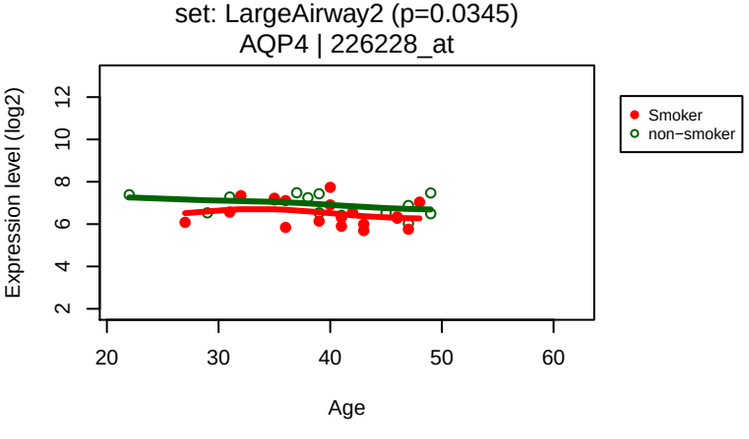

Gene expression ratio (smokers/nonsmokers)
gene: AQP4 | 226228_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2293
p=0.3081
p=1e−04
p=0.1578
p=0.0345
p=0.1832
p=0.0948
p=0.3087
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

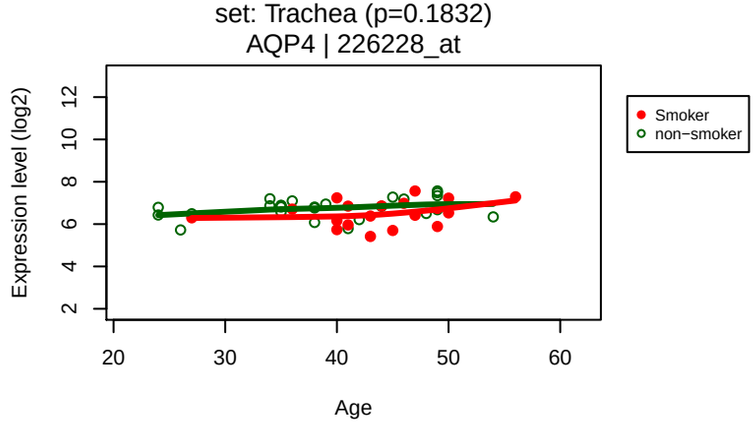

Gene expression ratio in Males
gene: AQP4 | 226228_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.808
p=0.0105
p=0.2991
p=0.0792
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP4 | 226228_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3253
p=0.8149
p=0.0965
p=0.7056
SmallAirway2
LargeAirway2
Trachea
Macrophage2

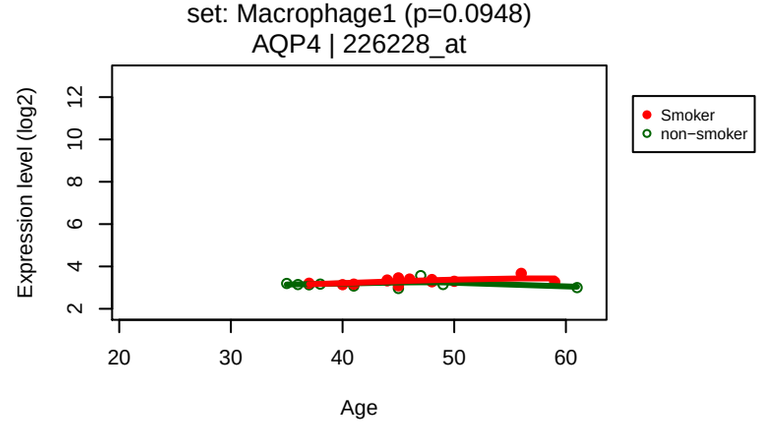

set: SmallAirway1 (p=0.2293)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker

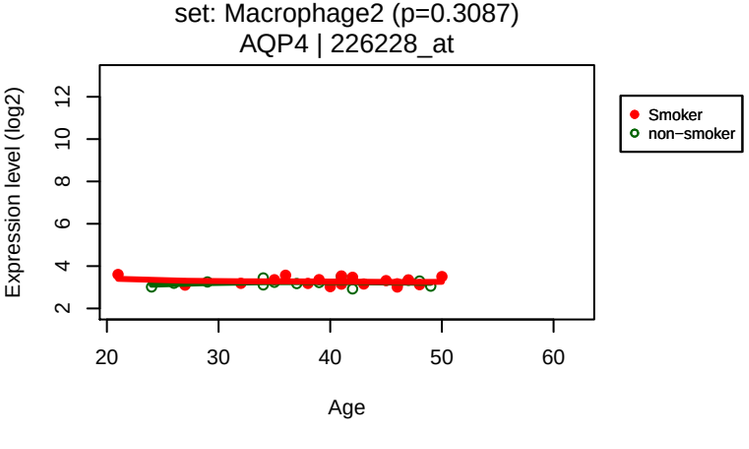

set: SmallAirway2 (p=0.3081)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker

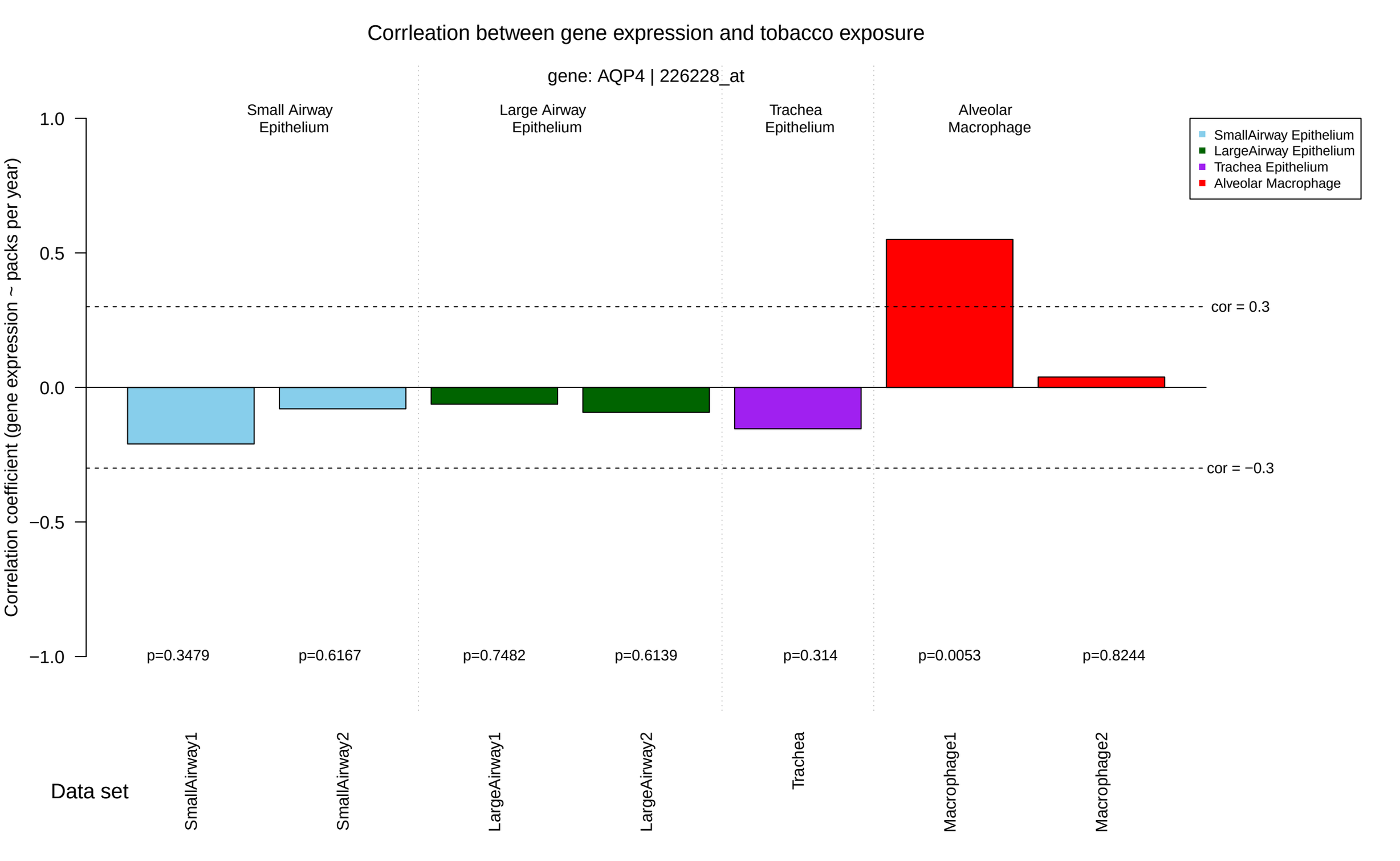

set: LargeAirway1 (p=0.1578)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0345)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1832)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0948)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3087)
AQP4 | 226228_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP4 | 226228_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3479
p=0.6167
p=0.7482
p=0.6139
p=0.314
p=0.0053
p=0.8244
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2